Abdullah Shamail

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Ames, Iowa | Open to Relocation

Education

Iowa State University — Ames, IA
Ph.D. Computer Engineering (GPA: 3.90/4.00) — *Expected May 2026*

- **Thesis:** Algorithms and techniques for analyzing moving clusters in spatio-temporal data
- **Coursework:** Machine Learning, NLP, Big Data Tools, Algorithms, Software Systems

Iowa State University — Ames, IA
M.S. Computer Engineering (GPA: 3.89/4.00) — *Dec 2024*

LUMS — Lahore, Pakistan
B.S. Computer Science (Dean's Honor List) — *May 2020*

Technical Skills

ML / AI: TensorFlow, PyTorch, GNNs, Transformers, RNNs, GANs, LLM, LangChain, RAG
Big Data & Systems: PySpark, Apache Spark, Flink, Hadoop, SQL, Linux/Unix
Languages: Python, C++, Java, JavaScript, PHP, SQL
Tools: LaTeX, Git, MDAnalysis, NumPy, Pandas, Docker, Azure/AWS, scikit-learn

Work Experience

Iowa State University — Ames, IA
Graduate Research Assistant — *Jan 2021 – Present*

- Created the **BARMC Algorithm (ACM SIGSPATIAL 2024)** by integrating spatial proximity with domain-specific semantic rules to detect molecular interactions - achieving **1.7× reduction** in computation time for **800,000-frame** (large-scale) simulations, enabling domain experts to precisely track Hydrogen-Bond persistence and molecular stability. *Awarded NSF Grant.*
- Designed a real-time system to automate the identification and visualization of long-lasting Hydrogen Bonds (HBs), generating detailed analytical reports that provide domain experts with deeper insights into molecular persistence. **(ACM SIGSPATIAL 2024)**
- Developed a domain-specific **Trajectory Compression (ADBIS 2022)** framework that achieved an **80% reduction in storage** without sacrificing query effectiveness. Optimized within-distance query processing to deliver **3× faster data retrieval**, allowing researchers to analyze massive simulation datasets with significantly lower hardware overhead. *Received Best Paper Award.*
- Designed **SDMC framework (Pending 2026)** implementing spatial indexing to achieve **17× to 65× speedups** in diversity-aware clustering using spatial-indexing and embedding-based similarity measures.
- Built pipelines for **motif mining** using geometric descriptors to identify aromatic ring-pair configurations for event prediction.
- Optimized Hadoop MapReduce algorithms for **Big Data Analysis**, achieving a **2× speedup** in text corpora processing.

Altair Engineering Inc. — Troy, MI
Machine Learning Engineer Intern — *May 2025 – Aug 2025*

- Engineered GNN and Transformer-based models for PhysicsAI models, achieving a **10× reduction in MAE** for large-scale simulations.
- Built and deployed scalable pipelines using PyTorch and TensorFlow for GPU-accelerated inference with Altair's production simulation platform.
- Optimized inference latency and memory profiling to support high-resolution physical predictions.

Iowa State University — Ames, IA
Student Instructor / Teaching Assistant — *Jan 2021 – Present*

- Mentored senior-year capstone teams (100+ students) through the full software development life-cycle and agile methodologies, where one led to a co-authored demo paper at **ACM SIGSPATIAL 2024**.
- Developed automated grading scripts for high-enrollment courses (200+ students) to provide rapid feedback on database and spreadsheet assignments.

Taar Consulting — Lahore, Pakistan
Software Engineer Intern — *May 2018 – June 2018*

- Implemented a full-stack Customer Feedback Plugin for Retail Pro Prism using Python and JavaScript
- Captured feedback across **1,000+ daily transactions** to drive customer experience analysis.

Technical Projects

AutoJobScout – AI Job Discovery Platform | ***LangGraph, LangChain, RAG, LLMs, ETL, Streamlit*** — 2025

- Architected an **Agentic AI system** using **LangGraph** to orchestrate multi-agent workflows and **RAG pipelines**, achieving **92% matching accuracy** via semantic search and LLM-powered resume analysis.
- Engineered an **ETL pipeline** across 7 asynchronous microservices to process **1,000+ job listings daily**, optimizing inference via **batch processing** to reduce latency to **<15 seconds**.

Selected ML Projects | ***PyTorch, Keras, TensorFlow*** — 2021 – 2022

- Built **LSTM** sarcasm classifiers (**84% precision**), **DCGAN** emoji synthesizers, and **CNNs** (CIFAR-10) for high-fidelity image classification.